CREATIVE COMPUTING LEVEL 5

Mobile App Development

Module Introduction

Mobile Market Share

Source: https://gs.statcounter.com/platform-market-share/desktop-mobile-tablet

Android Market Share (Mobile)

Android	iOS	Samsung
70.36%	29.25%	0.25%
Unknown	**KaiOS**	**Linux**
0.1%	0.02%	0.01%

Mobile Operating System Market Share Worldwide - January 2026

Source: https://gs.statcounter.com/os-market-share/mobile/worldwide

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Android Introduction

What Is Android?

- Android is a mobile operating system used to run smartphones, tablets, TVs, cars, and smart devices.
- It controls:
  - What apps can do
  - How devices use hardware (camera, GPS, touch)
  - How users interact with their device

Android

The Android Operating System

- Developed by Google
- Based on Linux
- Open-source
- Designed for touch-based devices

Android

Why Android?

- Large global audience
- Flexible customization
- Many device types
- Easier entry for beginners

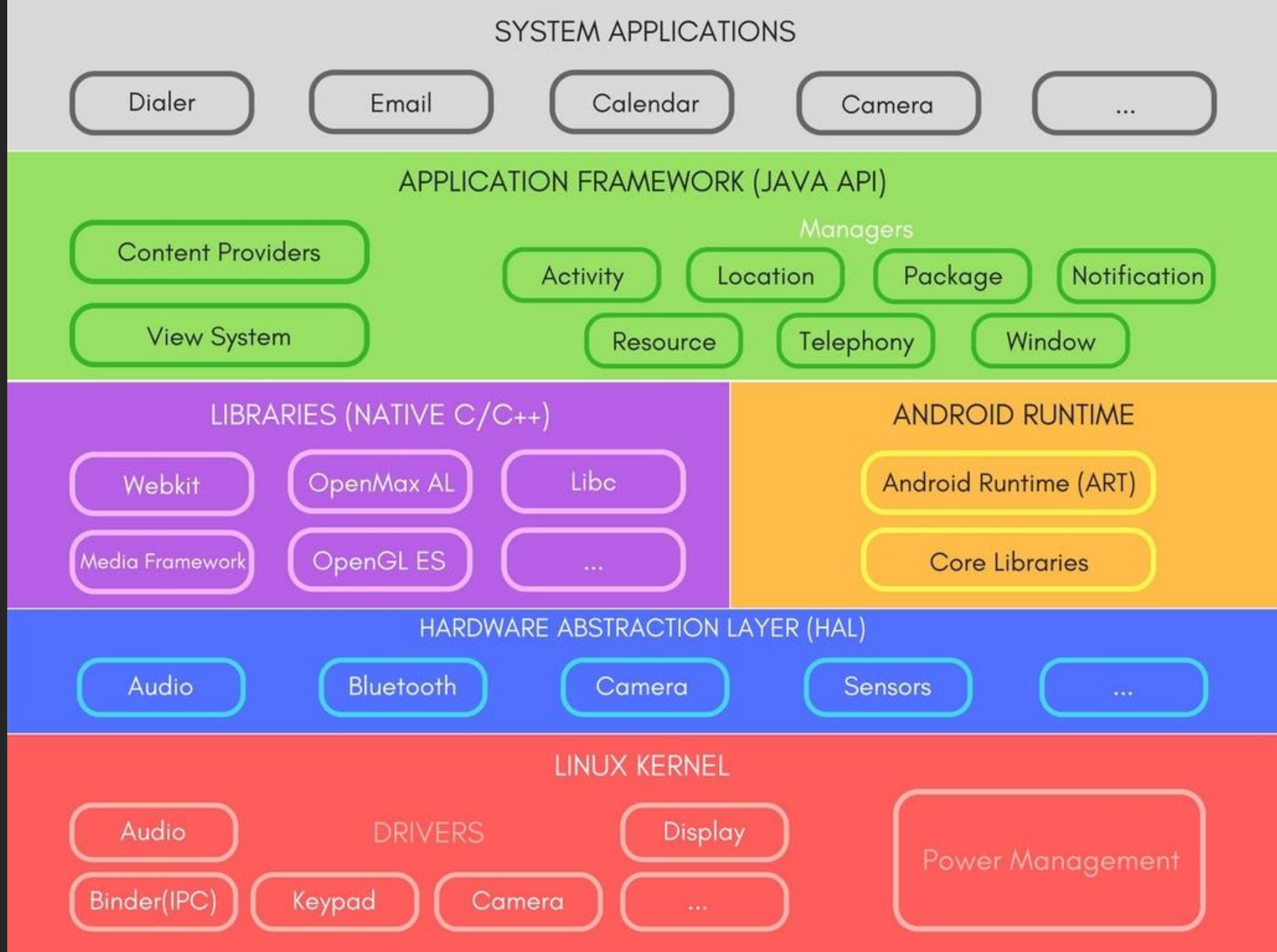
Android

Features of Android

- Touch gestures
- Multitasking
- Notifications
- App widgets
- Hardware integration

Android

Android Architecture



Android Versions

Android Versions

15.0	16.0	13.0
22.51%	15.84%	14.43%
14.0	12.0	11.0
13.9%	10.45%	8.44%

Mobile & Tablet Android Version Market Share Worldwide - January 2026

Source: https://gs.statcounter.com/android-version-market-share/mobile-tablet/worldwide

Android API Levels

- An API Level defines what features an app can use.
- Depends on:
  - Android version
  - Device capabilities

Android API Levels

Version	API level	Codename	Year 4
Android 16	Level 36	Baklava	2025
Android 15	Level 35	Vanilla Ice Cream	2024
Android 14	Level 34	Upside Down Cake	2023
Android 13	Level 33	Tiramisu	2022
…	…	…	…
Android 9	Level 28	Pie	2018

Android Applications

- An Android application (app) is software designed to perform a specific task on an Android device.
- Apps are created to:
  - Solve a problem
  - Provide a service
  - Entertain or inform users

Android Applications

Categories of Android Applications

Games	Social Media	Productivity
Education	Utilities	Entertainment

Android Ecosystem

- Devices (phones, tablets, TVs)
- Android operating system
- Apps
- Developers
- App stores

CREATIVE COMPUTING LEVEL 5

Mobile App Development

Android App Pipeline

Android App Pipeline

CREATIVE COMPUTING LEVEL 5

Mobile App Development

Android App Pipeline

What Makes a Good Mobile App?

- A good app:
  - Works correctly
  - Looks appealing
  - Is easy to use
  - Meets its purpose

FAAP Framework

- FAAP is a model for evaluating apps:
  - Functionality
  - Aesthetics
  - Architecture
  - Purpose

Functionality – “Does it actually work?”

- This is all about what the app can do.:
  - Touch gestures respond correctly
  - App loads quickly
  - Instructions or help available
  - No crashes or errors

Aesthetics – “Does it look and feel good?”

- This is the design side:
  - Consistency across screens
  - Appropriate use of white space
  - Graphics support content, not distract
  - Animations enhance experience, not annoy

Architecture – “Is it built properly?”

- This is the structure of the app:
  - Number of screens
  - Navigation menu clarity
  - Information hierarchy

Purpose – “Does it solve a real problem?”

- This is the why behind the app:
  - Who is it for?
  - What problem does it solve?
  - Why should anyone download it?
  - Does it make life easier, faster, or more fun?

Activity

Instructions:

- In groups, choose a popular app
- Rate the app 1-5 in each FAAP category
- Identify at least one strength and one weakness

Guide Questions:

- Functionality - *Does anything break or behave weirdly?*
- Aesthetic - *Is it pleasant, clear, and easy to use?*
- Architecture - *Does it load fast and feel smooth?*
- Purpose - *Is the goal of the app obvious and useful?*

Android Studio

- Official development tool
- Includes:
  - Code editor
  - Emulator
  - Debugging tools

Installing Android Studio

- Download Android Studio from the official website
  - developer.android.com/studio
- Install following on-screen instructions
- Open and explore the interface